Empirical Analysis of Transformers on Graph Connectivity

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1 Introduction

This report is a working summary of a sequence of experiments on a transformer trained to predict the connectivity matrix of an input graph. The starting point is the paper *Transformers Provably Learn Algorithmic Solutions for Graph Connectivity, But Only with the Right Data* (Ye, Fu, Jia, Sharan, 2026), which proves that for the simplified *Disentangled Transformer* architecture an L-layer model can solve connectivity on graphs of diameter at most 3^L, and that training the model exclusively on within-capacity graphs (the *data lever*) provably drives the model towards the algorithmic, matrix-powering solution rather than a degree-based heuristic. The motivating question of this report is whether the same data lever empirically helps a *standard* (non-disentangled) transformer in the same setting.

The experiments were run in the following order, and each step was decided as a reaction to what the previous step showed.

1. **Small-scale in-distribution checks (Chapter 3).** We first trained the standard transformer on Erdős–Rényi graphs with $n = 10$ and on the two structured families (2-chains and 2-cliques) at $n = 14$ used in the paper. The goal was only to verify that the model and the training pipeline behave reasonably in distribution and to get a baseline OOD picture. We deliberately do not draw strong conclusions from these experiments: the graphs are small and the in-distribution accuracy is essentially saturated, so the regime is not informative about the algorithm versus heuristic question.

2. **Diameter-controlled training on ER$(n = 40, p = 0.05)$ with the small model (Chapter 4).** We then moved to the setting that is actually relevant to the data lever: $n = 40$, $p = 0.05$, so that the three diameter cutoffs of interest $D \leq 7$, $D \leq 9$, $D \leq 11$ accept genuinely different fractions of graphs ($\sim 14\%, 55\%, 84\%$) and are statistically distinguishable. We trained four parallel runs (unfiltered, $D \leq 11$, $D \leq 9$, $D \leq 7$), kept the model architecture identical ($d_{\text{model}} = 64$, 1 head, 2 layers), and compared convergence speed, in-distribution accuracy, and OOD behaviour on 2-chains, 2-cliques, and unfiltered ER (broken down by graph diameter). The four small-model runs all settle at the same exact-match accuracy of about 0.80, which strongly suggested a model-capacity bottleneck rather than a data-distribution effect.

3. **Same diameter-controlled experiments with a larger model (Chapter 5).** To check the capacity hypothesis we re-ran the four diameter-controlled training experiments with a larger model ($d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, ~ 6.3M parameters), an online streaming data pipeline (so the model never sees the same graph twice), bf16 mixed precision, normalised-ReLU attention in place of softmax, and a 1,000,000-step training budget. We also added a 4-phase curriculum ($D \leq 7 \to D \leq 9 \to D \leq 10 \to D \leq 11$). With this model the in-distribution ceiling lifts and every run reaches $\geq 99.95\%$ exact match. We then re-evaluated all five trained models on the same OOD benchmark as Chapter 4.

Throughout the report we describe what we observe and try to avoid overinterpreting it. The data lever proposed by the paper turns out to be *useful for convergence speed* but, in our standard-transformer setup, *not useful for OOD generalisation*: across all the OOD tests we ran, the unfiltered model is at least as good as any of the diameter-restricted specialists, and on the harder OOD distributions (chains at large n_{active}, 2-cliques) all five large-model variants fail in qualitatively the same way. This is the main empirical finding of the report.

2 Model and task

We use a *Graph Connectivity Transformer* trained to predict the full connectivity matrix of an input graph.

Given an undirected graph with n nodes (the value of n varies across experiments and is specified in each chapter), the input is its adjacency matrix with self-loops added,

$$A \in \{0,1\}^{n \times n}.$$

The target is the connectivity matrix

$$R \in \{0,1\}^{n \times n}, \qquad R_{ij} = 1 \iff i \text{ and } j \text{ belong to the same connected component.}$$

Two model configurations are used across the report.

Small configuration (Chapters 3 and 4):

- Layers: 2, attention heads: 1, $d_{\text{model}} = 64$, $d_{\text{ff}} = 128$.
- Softmax attention, no dropout, batch size 256.
- Optimiser: AdamW, learning rate 10^{-3}, weight decay 10^{-2}, gradient clipping 1.0.
- Training steps: 500,000, evaluation every 1,000 steps.
- Fixed pre-generated training set of 6,000,000 graphs, fixed test set of 10,000 graphs.
- Parameter count: ~ 72k.

Large configuration (Chapter 5):

- Layers: 2, attention heads: 4, $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$.
- Normalised-ReLU attention ($\alpha = \frac{1}{n}\,\mathrm{ReLU}(QK^\top/\sqrt{d_h})$), no dropout, batch size 1,000.
- Optimiser: AdamW, peak learning rate 10^{-4} with linear warm-up and cosine decay, weight decay 10^{-4}, gradient clipping 1.0, bf16 mixed precision.
- Training steps: 1,000,000, evaluation every 5,000 steps on a pre-generated test set of 10,000 graphs.
- Training data is generated *on the fly* via rejection sampling: the model never sees the same training graph twice (effective training set size $\approx 10^9$ graphs).
- Parameter count: 6,347,304.

In every experiment we track the following metrics on the held-out test set:

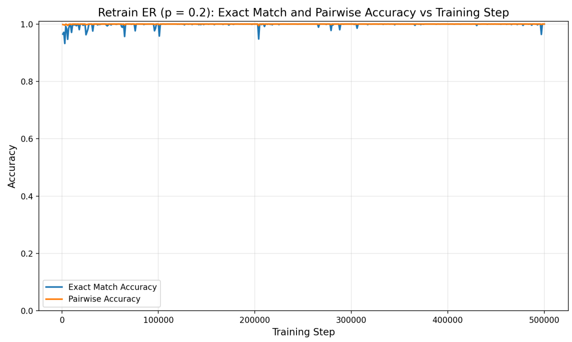
- **Exact-match accuracy:** fraction of test graphs for which the predicted connectivity matrix is exactly correct.

- **Pairwise accuracy:** average accuracy across all ordered node pairs in the test set.
- **Pairwise accuracy by distance:** pairwise accuracy conditioned on the shortest-path distance between the two nodes in the test graph.
- **Training loss:** binary cross-entropy loss.

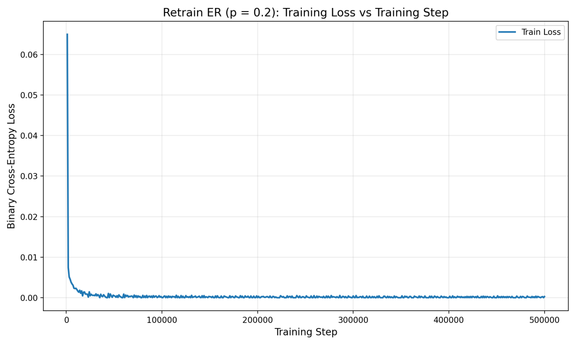
3 Chapter 1 — Small-scale experiments ($n = 10$ and $n = 14$)

The purpose of this chapter is to document the warm-up experiments. The model is the small configuration of Section 2, with the training sets and graph sizes specified in each subsection. As argued in the introduction, these experiments are not designed to discriminate between an algorithmic and a heuristic solution: the graphs are small, the in-distribution accuracy saturates very quickly, and the only OOD test set we use here (ER($n = 14$) at three densities) is a structurally simple shift. We therefore restrict the discussion to a brief description of each run and a short note on what the figures show.

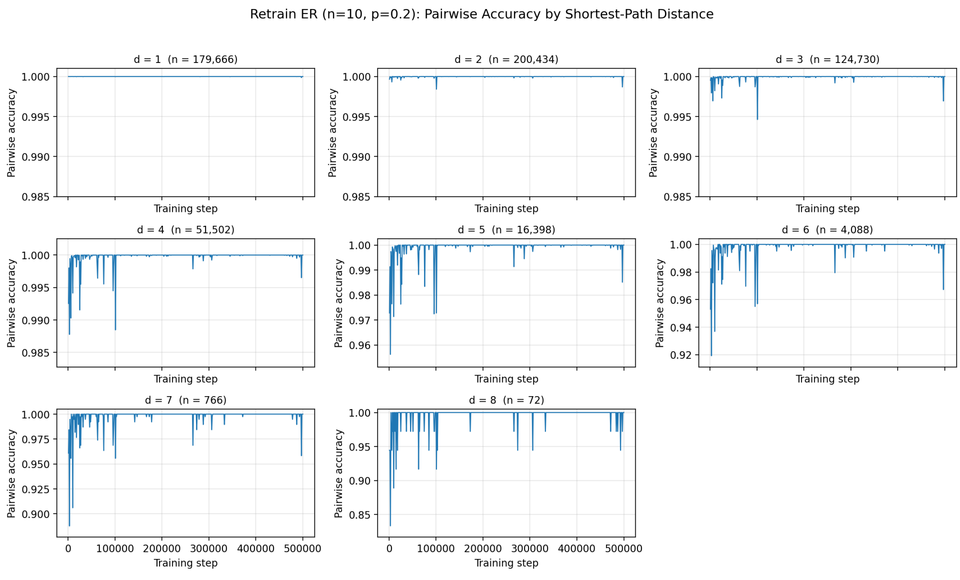
3.1 ER($n = 10$)



(a) Exact match and pairwise accuracy ($p = 0.2$).

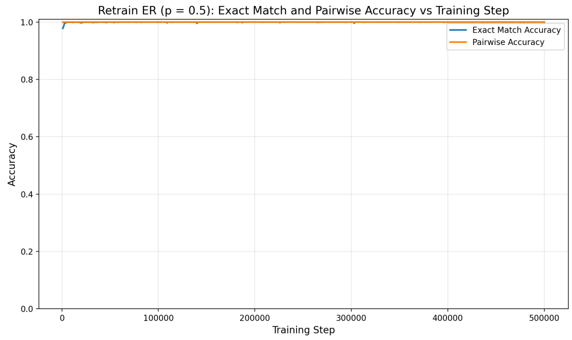


(b) Training loss ($p = 0.2$).

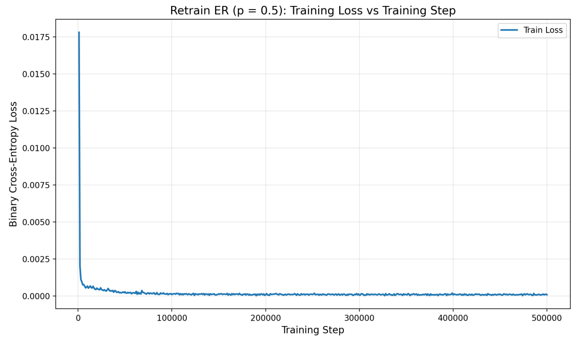


(c) Pairwise accuracy by shortest-path distance ($p = 0.2$). The number above each panel is the count of test pairs at that distance.

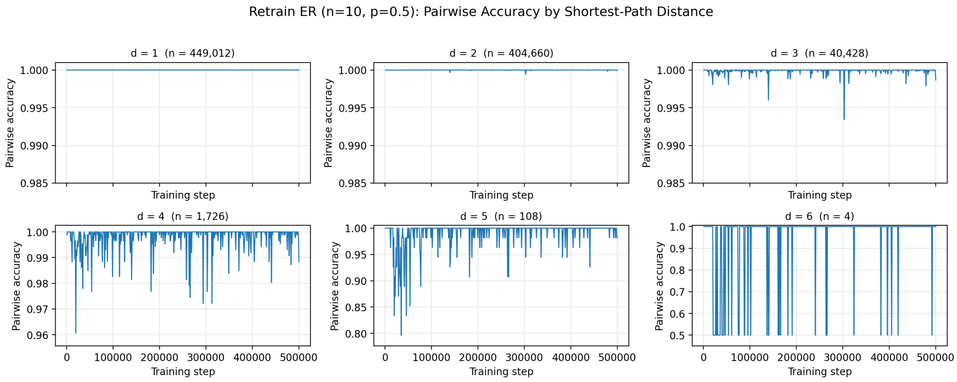
Figure 1: ER($n = 10$, $p = 0.2$) training run.



(a) Exact match and pairwise accuracy ($p = 0.5$).



(b) Training loss ($p = 0.5$).

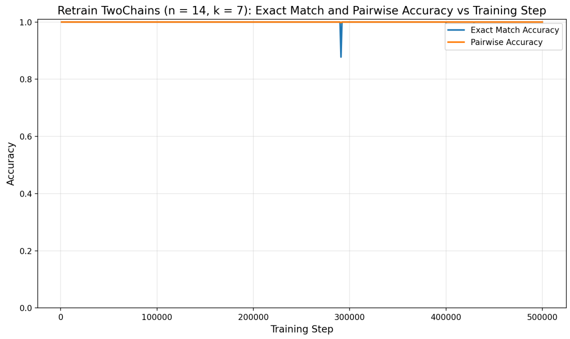


(c) Pairwise accuracy by shortest-path distance ($p = 0.5$). The number above each panel is the count of test pairs at that distance.

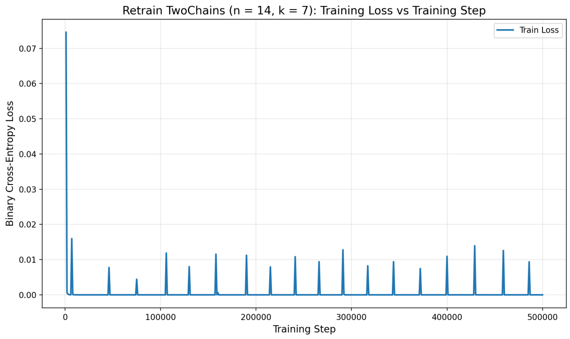
Figure 2: ER($n = 10$, $p = 0.5$) training run.

What the figures show. Both runs saturate within the first few thousand steps. The loss curves are very small but show a slight residual oscillation. Per-distance accuracy is essentially perfect up to the largest distance reasonably represented in the test set; the few visible drops at $d \geq 6$ are on bins that contain very few samples.

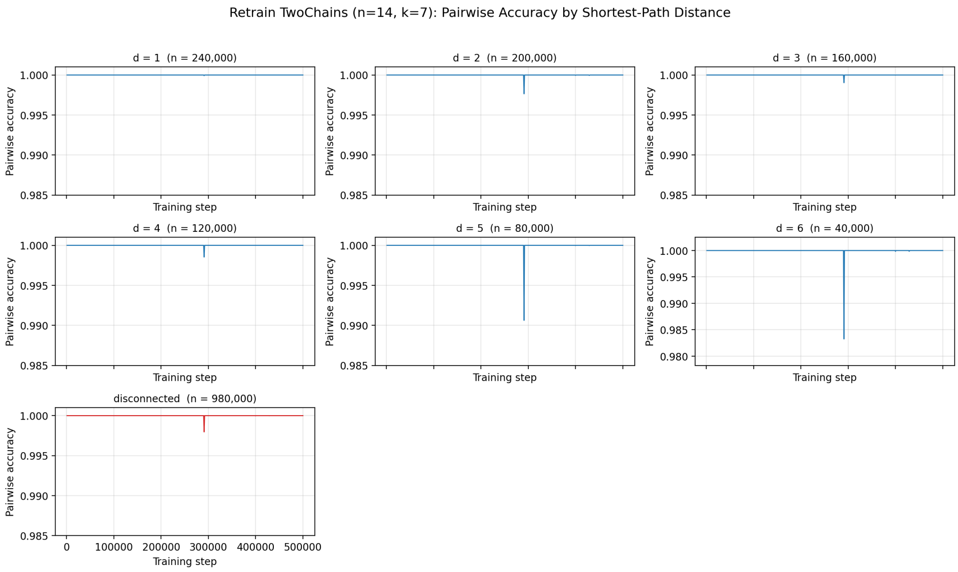
3.2 2-chains ($n = 14$)



(a) Exact match and pairwise accuracy.



(b) Training loss.



(c) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

Figure 3: 2-chains ($n = 14, k = 7$) training run.

What the figures show. The model reaches near-perfect accuracy very quickly. The training loss exhibits a clear periodic spike pattern which we discuss next.

3.2.1 Periodic spikes in the training loss

The training-loss curve shows periodic spikes of magnitude ~ 0.01–0.015, appearing roughly every 25,000–30,000 steps. The period matches the epoch length: with 6,000,000 training graphs and batch size 256, one epoch is

$$T_{\text{epoch}} = \frac{6{,}000{,}000}{256} \approx 23{,}438$$

training steps. At each epoch boundary the dataset is reshuffled. Because AdamW maintains running estimates of the first and second moments of the gradients calibrated on the previous local sequence of minibatches, the sudden change in mini-batch order produces a short transient mismatch between the optimiser state and the new local gradient statistics; after a few hundred steps the optimiser re-adapts and the loss returns to its baseline. The observed period is slightly longer than T_{epoch} because the displayed loss is a 1,000-step moving average.

3.2.2 Out-of-distribution evaluation (chains → ER)

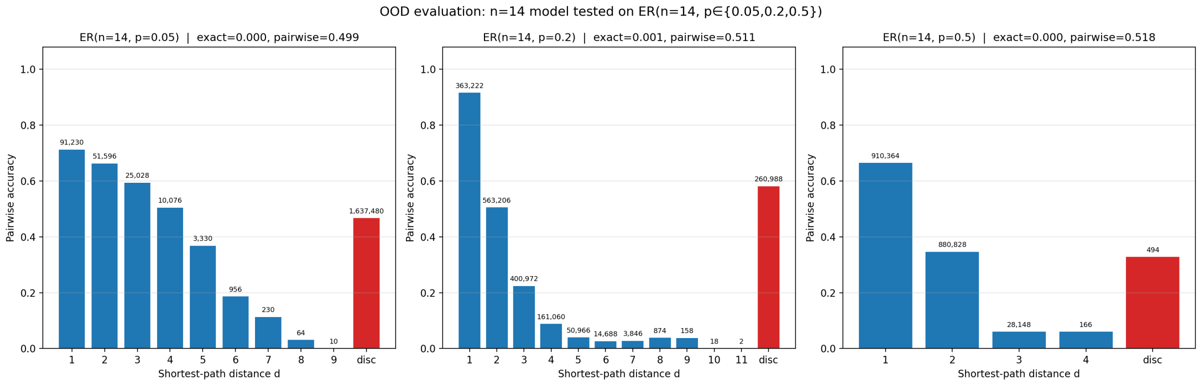
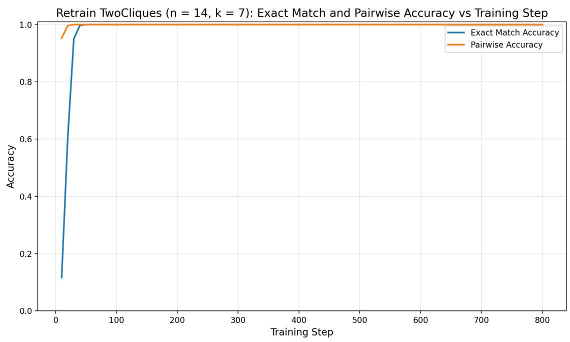


Figure 4: 2-chains ($n = 14$) model evaluated on ER($n = 14, p$) for $p \in \{0.05, 0.2, 0.5\}$. Per-distance pairwise accuracy.

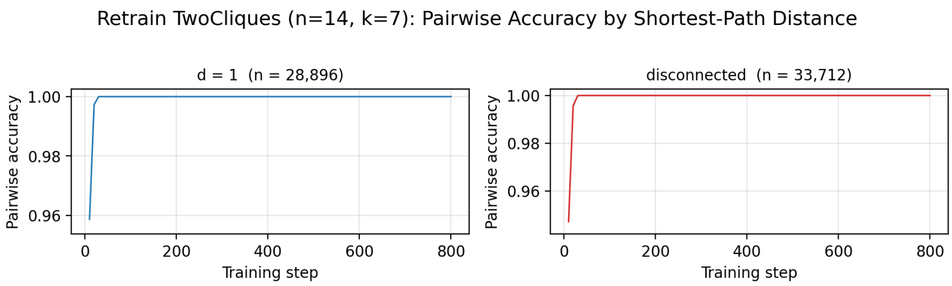
Exact-match accuracy is ≈ 0 at all three densities and pairwise accuracy lies in 0.50–0.52. The per-distance breakdown shows that accuracy is highest at $d = 1, 2$ and decays for larger distances; the "disconnected" bin is the largest source of error.

3.3 2-cliques ($n = 14$)



(a) Exact match and pairwise accuracy.

(b) Training loss.



(c) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

Figure 5: 2-cliques ($n = 14, k = 7$) training run.

What the figures show. The 2-cliques family with $n = 14$ and $k = 7$ has only $\binom{14}{7}/2 = 1716$ distinct graphs; we therefore use the exhaustive set split 80%/20% and cap training at 800 steps. The model saturates within a few hundred steps; in-distribution accuracy on this dataset is mainly evidence of memorisation rather than of generalisation.

3.3.1 Out-of-distribution evaluation (cliques → ER)

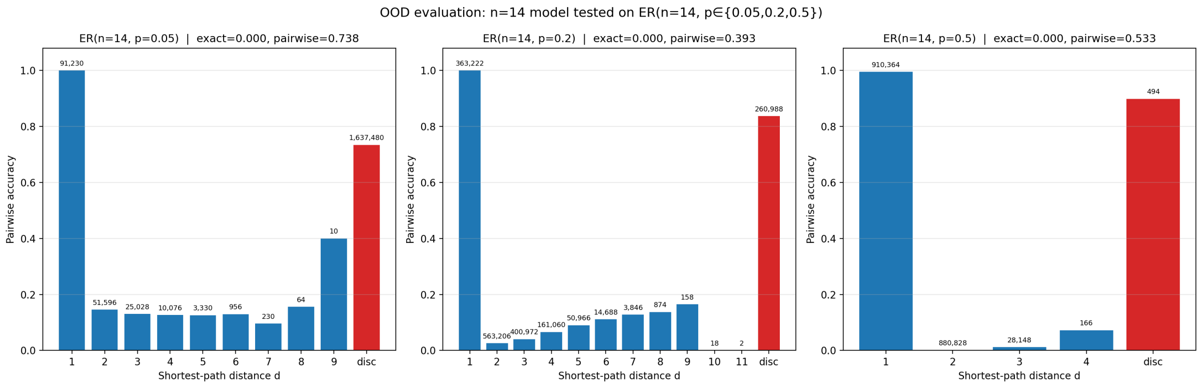


Figure 6: 2-cliques ($n = 14, k = 7$) model evaluated on ER($n = 14, p$) for $p \in \{0.05, 0.2, 0.5\}$. Per-distance pairwise accuracy.

Exact match is 0% at all three densities. The per-distance breakdown is informative: the model gets $d = 1$ pairs essentially right ($\geq$0.99 at $p = 0.2, 0.5$ and 1.000 at $p = 0.05$) and collapses to near-zero accuracy for $d \geq 2$. Effectively the trained model has learnt the rule

$$\hat{R}_{ij} = 1 \iff i \text{ and } j \text{ share a direct edge in } A,$$

which is the correct answer inside a clique but not on a generic graph.

4 Chapter 2 — Diameter-controlled training on ER($n = 40$) (small model)

4.1 Motivation and setup

The small-scale experiments of Chapter 3 are not informative about the question we ultimately care about, namely whether the *data lever* of the disentangled-transformer paper transfers to a standard transformer. To probe that question we need a regime in which (i) the model has to do non-trivial multi-hop reasoning, (ii) the diameter cut-offs $D \leq 7$, $D \leq 9$, $D \leq 11$ produce *visibly different* training distributions, and (iii) the model is still small enough that we can train four parallel runs to convergence.

The choice $n = 40$, $p = 0.05$ (expected degree $np = 2$) makes the diameter cut-offs statistically meaningful: at this density the three cut-offs accept respectively $\approx$ 14%, 55%, and 84% of randomly sampled graphs. With smaller n (e.g. $n = 20$) all three cut-offs accept more than 95% of graphs and the comparison becomes degenerate.

We run four parallel experiments with the small configuration of Section 2, identical except for the diameter filter applied to the training graphs:

- **Unfiltered:** ER$(40, 0.05)$.
- **$D \leq 11$:** ER$(40, 0.05)$ rejecting any graph whose largest finite shortest-path distance exceeds 11.

- **$D \le 9$:** same with cut-off 9. This is the *predicted optimum* for a 2-layer model, since $3^L = 9$.
- **$D \le 7$:** same with cut-off 7.

Each run uses 6,000,000 training graphs, 10,000 in-distribution test graphs (drawn from the same filtered distribution), and 500,000 optimisation steps.

4.2 Cross-experiment view: convergence and final accuracy

Table 1 summarises the four runs. The most striking effect is on convergence speed: tighter diameter constraints lead to much faster learning, with the most restrictive cut-off ($D \le 7$) roughly 17× faster than the unfiltered run on the "99%-pairwise-accuracy" milestone.

Experiment	Best exact match	Final pairwise	Final loss	Steps to 99% pairwise
Unfiltered	0.812	0.990	0.0260	273,000
$D \le 11$	0.808	0.993	0.0193	78,000
$D \le 9$	0.802	0.995	0.0135	38,000
$D \le 7$	**0.823**	0.998	0.0060	**16,000**

Table 1: Summary of the four diameter-controlled small-model experiments on ER($n = 40, p = 0.05$). All runs use 6,000,000 training graphs and 500,000 optimisation steps.

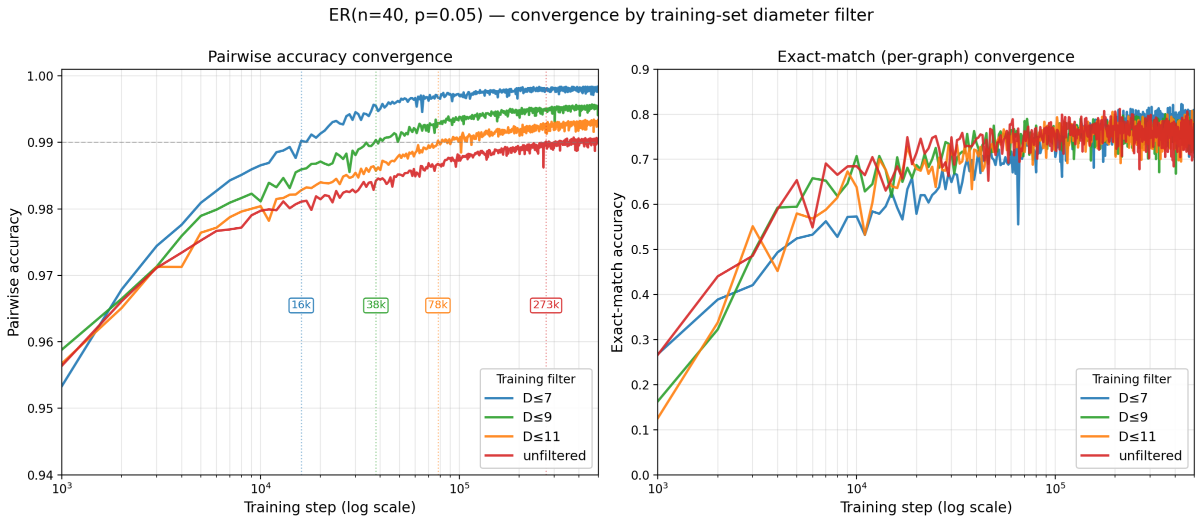


Figure 7: Convergence comparison across the four diameter-controlled experiments. *Left:* pairwise accuracy as a function of the training step (log scale); boxed labels mark the step at which each model first crosses the 99% threshold (dashed line). *Right:* per-graph exact-match accuracy on the same log-scaled axis. Despite the large gap in convergence speed, all four runs settle at very similar final values.

Convergence speed is monotone in the diameter filter. Tighter constraints lead to faster learning. This is consistent with the picture proposed by the disentangled-transformer paper, in which an L-layer model is expected to learn most efficiently on graphs whose diameter does not exceed 3^L.

Final accuracy is essentially unaffected. All four runs settle in a narrow band around 80% exact match and 99% pairwise accuracy. With $n = 40$ each test graph contains $40 \times 39 = 1{,}560$ off-diagonal pairs, so a pairwise accuracy of 0.99 corresponds on average to about 15 wrong pairs per graph, which is enough to drive the exact-match metric down. The fact that all four runs land at roughly 0.80 exact match strongly suggests that the bottleneck of the present setup is the size of the model ($d_{\text{model}} = 64$, one head, two layers) rather than the structure of the training data; this is the direct motivation for the large-model experiments of Chapter 5.

4.3 Cross-experiment view: pairwise accuracy by distance

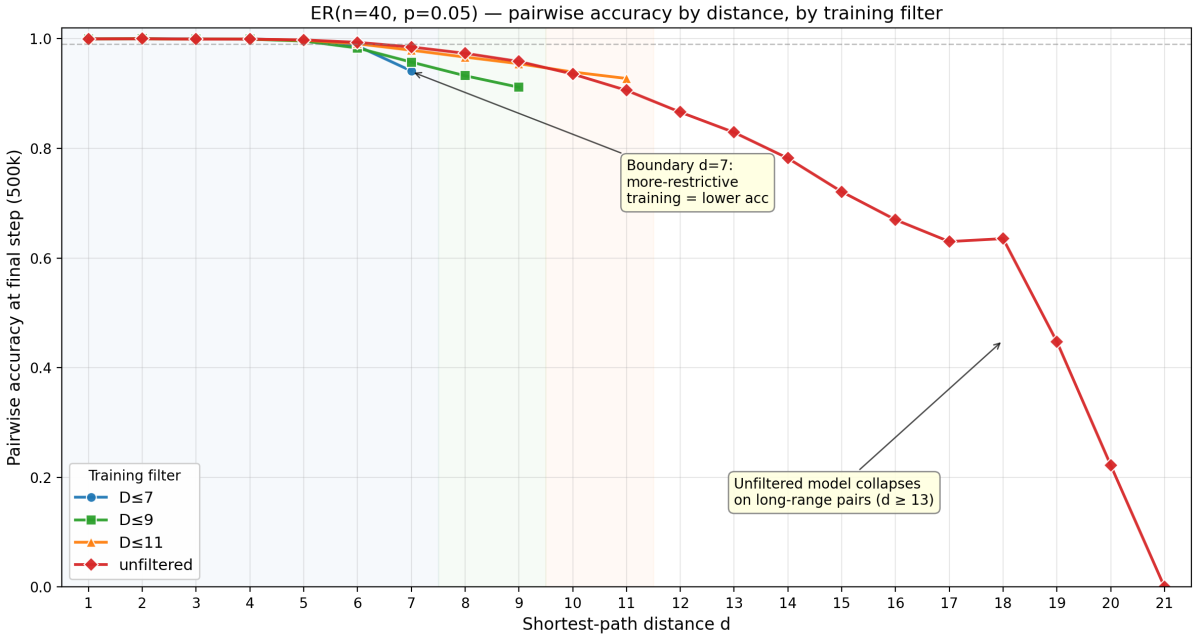


Figure 8: Final pairwise accuracy broken down by shortest-path distance d for the four diameter-controlled experiments. Background shading indicates the support of each filtered training distribution. Each curve is truncated at the maximum d present in the corresponding test set.

Three observations stand out.

Accuracy is near-perfect at small distances. For $d \leq 5$ all four models predict pairs correctly with probability greater than 0.995 and the four curves are essentially indistinguishable.

Restrictive training is worse at the boundary. At $d = 7$ the ranking of the four models is the opposite of what one might expect: unfiltered reaches 0.985, $D \leq 11$ reaches 0.979, $D \leq 9$ reaches 0.957, and $D \leq 7$ reaches only 0.941.

The unfiltered model decays slowly on long-range pairs. Beyond $d = 11$, only the unfiltered model has test data on which to be evaluated; its accuracy drops to 0.83 at $d = 13$, 0.45 at $d = 19$ and 0 at $d = 21$.

4.4 Cross-experiment view: out-of-distribution

The same four models, evaluated on three OOD test sets (10,000 graphs each):

- **2-chains** with $n_{\text{active}} \in \{14, 16, \ldots, 40\}$, padded to 40×40 with isolated nodes;

- **2-cliques** with the same n_{active} protocol;
- **Unfiltered** $\text{ER}(40, 0.05)$ with the per-graph-diameter breakdown of exact-match accuracy (for the unfiltered model this coincides with the in-distribution evaluation and is reported for direct comparison).

2-chains

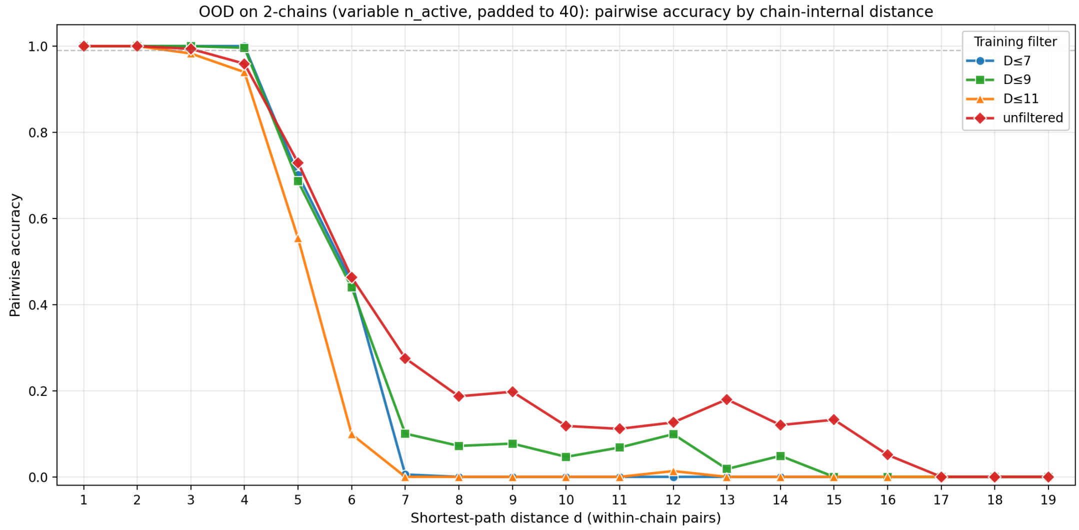


Figure 9: 2-chains OOD: pairwise accuracy as a function of the chain-internal distance d, compared across the four diameter-controlled small-model runs. Each curve is averaged over all n_{active} values.

Exact match is 0 for all four models. Average pairwise accuracy on the ~ 3.7M chain-internal reachable pairs is:

- Unfiltered: 0.636
- $D \leq 9$: 0.597
- $D \leq 7$: 0.577
- $D \leq 11$: 0.526

The unfiltered model is the most robust on chain transfer despite never having been trained on a chain. The diameter-restricted models drop to zero accuracy at, or even before, their respective training cut-offs.

2-cliques

Three of the four models reach exact match 0. The only exception is the $D \leq 7$ model, which scores 0.068 overall: this signal comes entirely from the $n_{\text{active}} = 14$ subset (two 7-cliques padded with 26 isolated nodes) where the model achieves 100% exact match, and is 0 for every other n_{active}. Per-distance accuracy across all four models is essentially the same as in the small-scale 2-cliques OOD test (Chapter 3): $d = 1$ accuracy is very low (≤ 0.08) and the "disconnected" bin dominates the pairwise number.

Unfiltered ER, by graph diameter

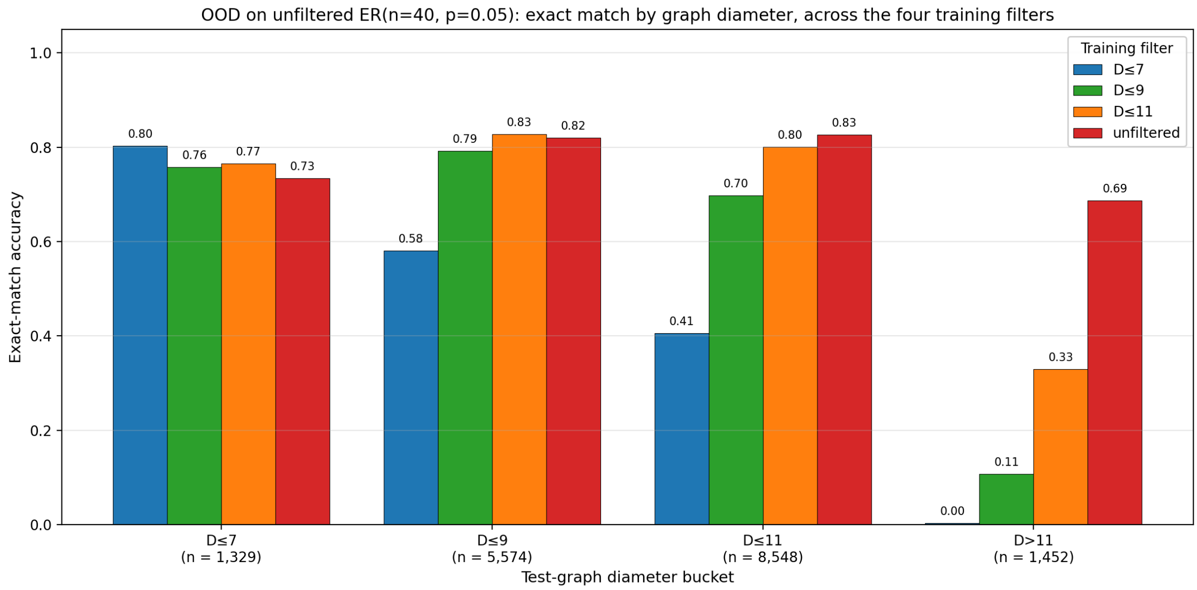


Figure 10: OOD evaluation on unfiltered ER$(40, 0.05)$: exact-match accuracy broken down by the diameter of the test graph, compared across the four diameter-controlled small-model runs.

Model	$D \leq 7$	$D \leq 9$	$D \leq 11$	$D > 11$	overall
Unfiltered	0.734	0.820	0.826	**0.687**	0.806
$D \leq 11$	0.765	0.828	0.800	0.329	0.732
$D \leq 9$	0.758	0.792	0.697	0.107	0.612
$D \leq 7$	**0.803**	0.580	0.405	0.003	0.347

Table 2: Exact-match accuracy on a fixed test set of 10,000 unfiltered ER$(40, 0.05)$ graphs, broken down by the test-graph diameter (number of graphs per bucket: 1,329, 5,574, 8,548, 1,452). Bold entries mark the best model in each column.

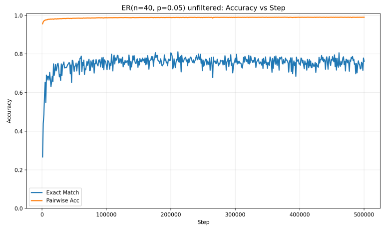
Three patterns:

- Each diameter-restricted model wins on the bucket that matches its training distribution and the unfiltered model wins on the tail bucket $D > 11$.
- The tighter the filter, the more catastrophic the failure on the excluded bucket: 0.329 for $D \leq 11$, 0.107 for $D \leq 9$, 0.003 for $D \leq 7$.
- Even on buckets nominally inside every model's training distribution (e.g. $D \leq 7$, $D \leq 9$), the unfiltered model is competitive with or better than the corresponding specialist.

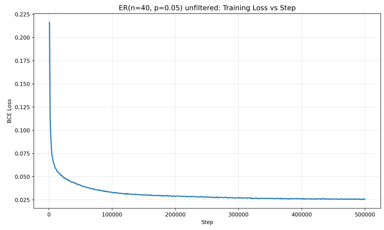
4.5 Per-experiment figures (small model)

The following four subsections collect, for each filter, the in-distribution training curves and the per-experiment OOD plots that were aggregated in the cross-experiment view above. The text is intentionally short.

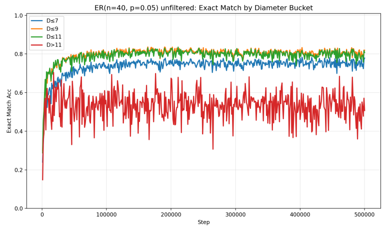
4.5.1 Unfiltered



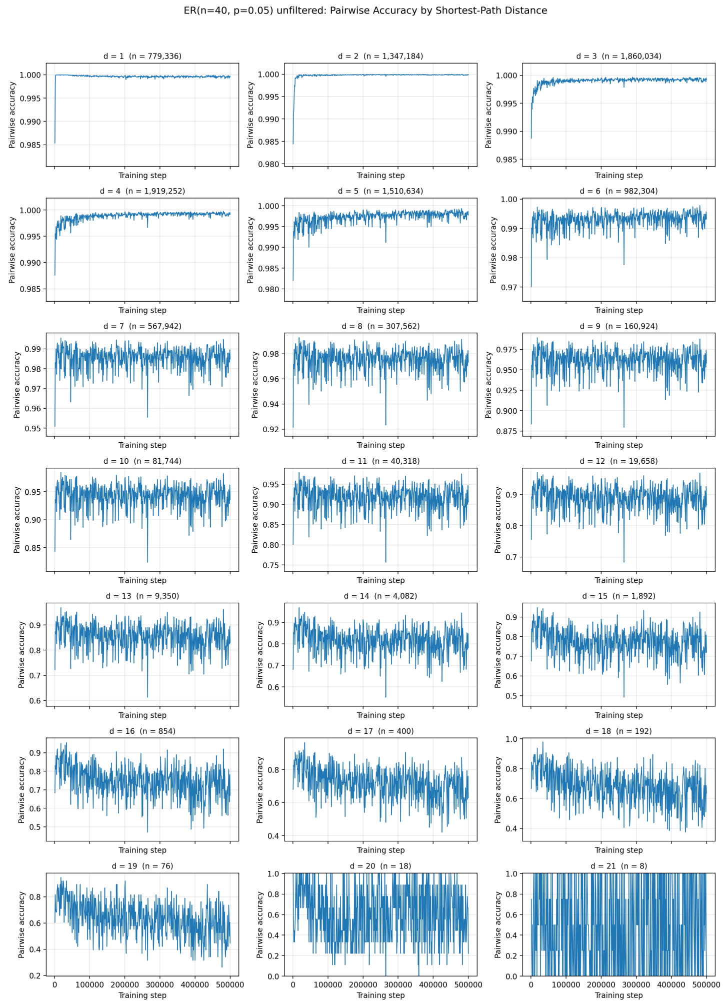
(a) Exact match and pairwise accuracy.



(b) Training loss.

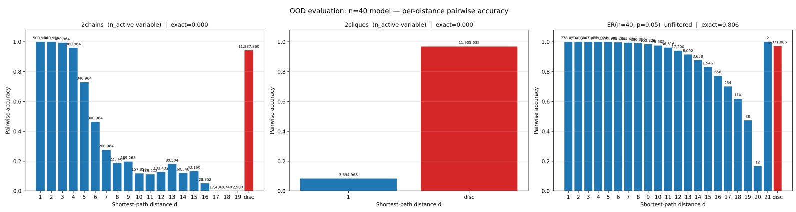


(c) Exact match per test-graph diameter bucket.

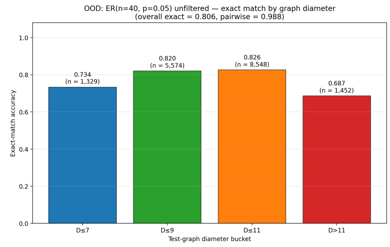


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

Figure 11: Unfiltered, in distribution.



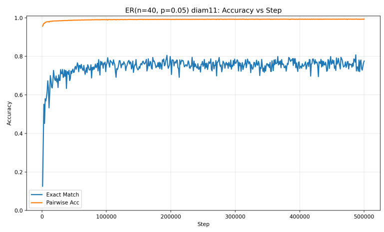
(a) Per-distance pairwise accuracy on the three OOD sets.



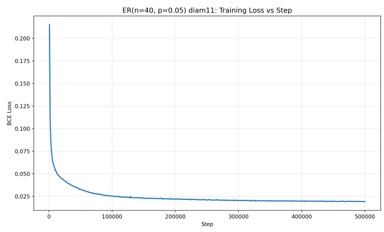
(b) Exact match on unfiltered ER by diameter bucket.

Figure 12: Unfiltered, OOD.

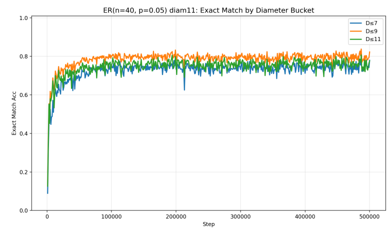
4.5.2 $D \leq 11$



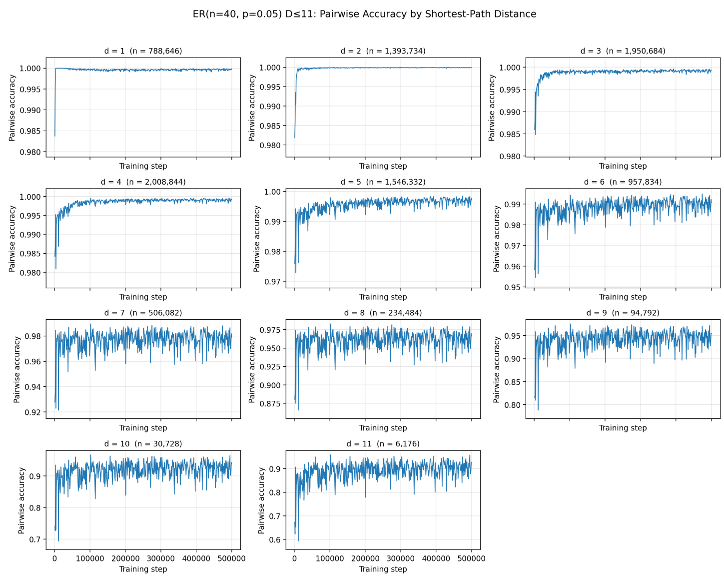
(a) Exact match and pairwise accuracy.



(b) Training loss.

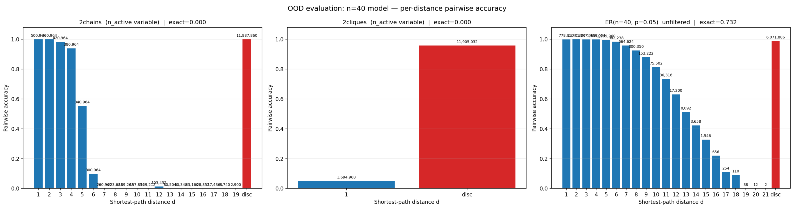


(c) Exact match per test-graph diameter bucket.

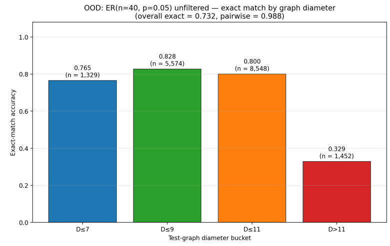


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

Figure 13: $D \leq 11$, in distribution.



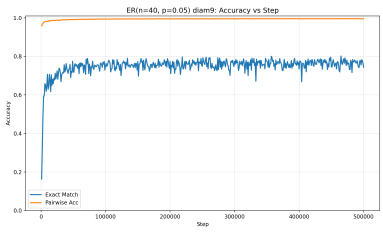
(a) Per-distance pairwise accuracy on the three OOD sets.



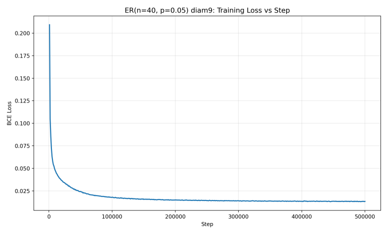
(b) Exact match on unfiltered ER by diameter bucket.

Figure 14: $D \leq 11$, OOD.

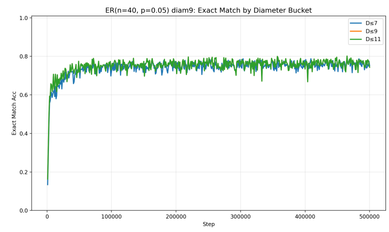
4.5.3 $D \leq 9$



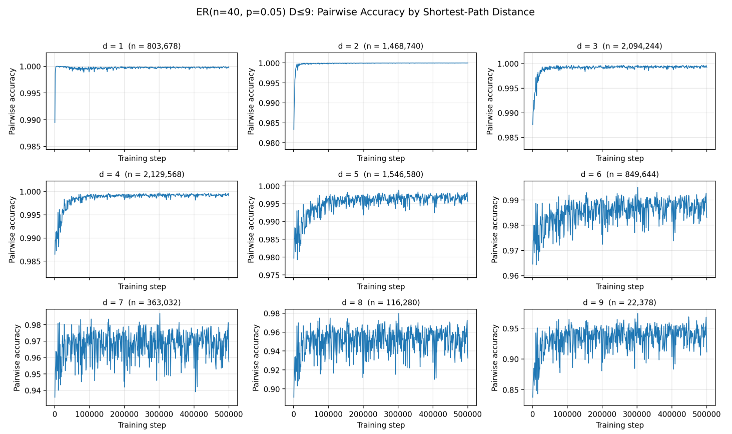
(a) Exact match and pairwise accuracy.



(b) Training loss.

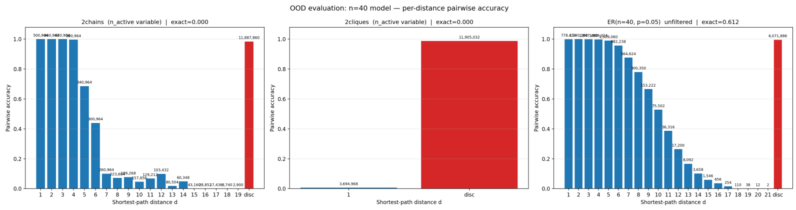


(c) Exact match per test-graph diameter bucket.

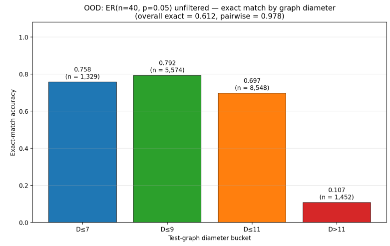


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

Figure 15: $D \leq 9$, in distribution.



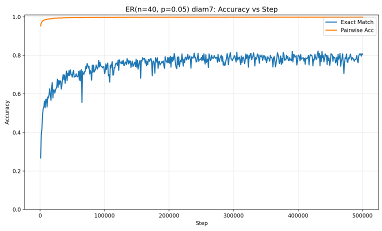
(a) Per-distance pairwise accuracy on the three OOD sets.



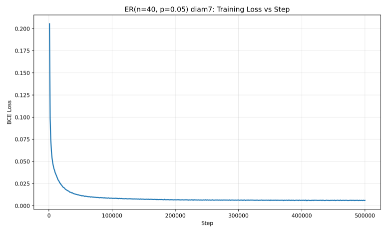
(b) Exact match on unfiltered ER by diameter bucket.

Figure 16: $D \leq 9$, OOD.

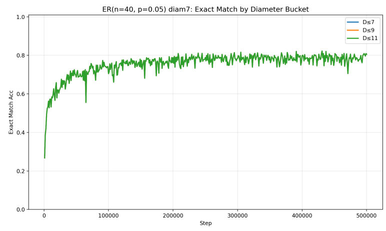
4.5.4 $D \leq 7$



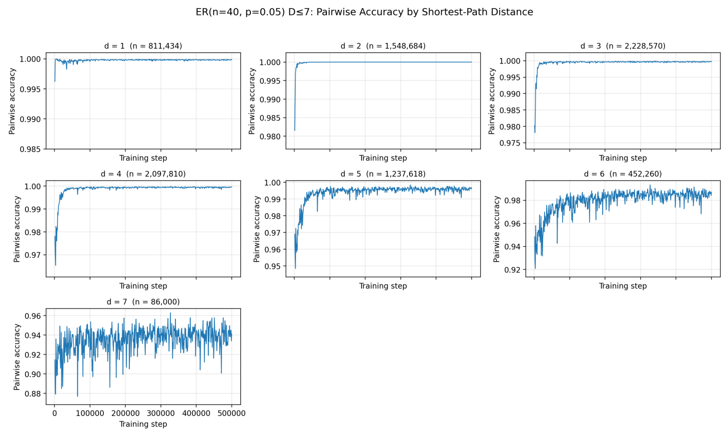
(a) Exact match and pairwise accuracy.



(b) Training loss.

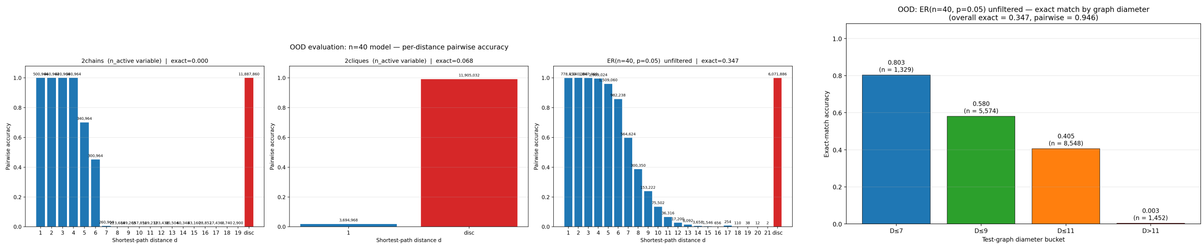


(c) Exact match per test-graph diameter bucket.



(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

Figure 17: $D \leq 7$, in distribution.



(a) Per-distance pairwise accuracy on the three OOD sets.

(b) Exact match on unfiltered ER by diameter bucket.

Figure 18: $D \leq 7$, OOD.

4.6 Comments on Chapter 4

The four diameter-controlled small-model runs reach essentially the same in-distribution exact-match accuracy (≈ 0.80) but converge at very different speeds. The diameter filter helps the model *learn faster*, but the final achievable accuracy does not depend on the filter in this setup. The fact that all four runs converge to the same accuracy is the main reason we then re-ran the same protocol with a larger model in Chapter 5: if the bottleneck is model size rather than training distribution, the data lever effect should become more visible once the model is large enough to actually fit the easier filtered distributions.

On the OOD side, the picture is already mostly negative for the data lever in the standard-transformer setting. The unfiltered model is the most robust on 2-chains transfer, no model gets any 2-cliques graph exactly right at $n_{\text{active}} > 14$, and on the unfiltered-ER per-bucket evaluation the unfiltered model is competitive with each diameter specialist on the specialist's bucket and is strictly better on the $D > 11$ bucket. The $D \leq 9$ configuration that the disentangled theory predicts as the empirical optimum for $L = 2$ is, by every metric we measure, not the best of the four.

5 Chapter 3 — Diameter-controlled training on ER$(n = 40)$ (large model)

5.1 Motivation and setup

The Chapter 4 experiments left a clean motivating question. The four small-model runs converge at very different speeds but settle at the same in-distribution exact-match accuracy of about 0.80, which is consistent with a model-size rather than data-distribution bottleneck. Two follow-up actions are therefore natural: increase the model capacity to lift this ceiling, and re-run the diameter-controlled comparison in the regime where every run reaches near-perfect in-distribution accuracy, so that we can read OOD performance without the in-distribution number acting as a confound.

We use the large configuration of Section 2: $L = 2$ attention layers, $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, 4 attention heads, normalised-ReLU attention, bf16 mixed precision, online streaming of training graphs (each training graph is freshly sampled by rejection sampling at each step, so the model never sees the same graph twice), batch size 1,000 and a training budget of 1,000,000 optimisation steps. We additionally include a 4-phase *curriculum* run that starts on $D \leq 7$, advances to $D \leq 9$, $D \leq 10$ and finally $D \leq 11$, with each phase boundary triggered by the smoothed training loss falling below a pre-set threshold; the combined budget is again 1,000,000 steps. All five runs share the same optimiser, model size, and OOD evaluation protocol.

5.2 Cross-experiment view: convergence and final accuracy

Table 3 reports best in-distribution exact match, convergence steps to the 0.99 exact-match milestone, and a few finer milestones for the five large-model runs.

Experiment	Best exact match	step @ EM ≥ 0.99	step @ EM ≥ 0.999	step @ EM ≥ 0.9999
Unfiltered	0.9997	180,000	570,000	—
$D \leq 11$	0.9998	120,000	535,000	—
Curriculum	0.9998	110,000	485,000	—
$D \leq 9$	1.0000	85,000	330,000	500,000
$D \leq 7$	1.0000	**10,000**	**85,000**	**170,000**

Table 3: Convergence summary for the five large-model runs. All runs use online streaming and 1,000,000 optimisation steps. "Step @ EM $\geq x$" is the first evaluation step at which exact-match accuracy on the held-out test set reaches the threshold x. A dash indicates that the threshold was not reached during training.

The convergence ordering is the same one already observed for the small model in Chapter 4: convergence speed is monotone in the diameter filter, with $D \leq 7$ the fastest and unfiltered the slowest. Two differences with respect to Chapter 4 are worth noting:

- Every large-model run reaches ≥ 0.9995 in-distribution exact match; the 0.80 ceiling of the small model is gone.
- The curriculum run lands essentially on the $D \leq 11$ run. A glance at the run's phase log makes this obvious: the loss thresholds for phases 1–3 are met within 30,000 steps total, so the curriculum spends about 970,000 of its 1,000,000 steps in the final $D \leq 11$ phase. The curriculum as configured is therefore close to "$D \leq 11$ with a short $D \leq 7$ warm-up".

5.3 Cross-experiment view: out-of-distribution

We re-evaluate each of the five large-model runs on:

- **2-chains** with $n_{\text{active}} \in \{14, 16, \ldots, 40\}$, padded to 40×40;
- **2-cliques** with the same protocol;
- **Unfiltered ER**$(40, 0.05)$, with the per-graph-diameter breakdown.

Each test set contains 10,000 graphs.

2-chains: exact-match cliff by n_{active}

n_{active}	Unfiltered	$D \leq 11$	$D \leq 9$	$D \leq 7$	Curriculum
$14, 16$	1.00	1.00	1.00	1.00	1.00
$18, 20$	1.00	1.00	1.00	**0.00**	1.00
≥ 22	0.00	0.00	0.00	0.00	0.00

Table 4: Exact-match accuracy on the 2-chains OOD set as a function of n_{active}, grouped over rows with identical behaviour. Each n_{active} bin contains ≈ 700 graphs.

Four of the five models (unfiltered, $D \leq 11$, $D \leq 9$, curriculum) exhibit the same step function: exact match 1.0 up to $n_{\text{active}} = 20$, then 0.0 from $n_{\text{active}} = 22$ onward. The $D \leq 7$ model loses two additional bins ($n_{\text{active}} = 18, 20$). In other words, on 2-chains the data lever does not help in the large-model regime either: the unfiltered model and the three filtered models are indistinguishable from each other, with the only effect of the tightest filter ($D \leq 7$) being to move the cliff earlier.

2-cliques

All five large-model runs reach exact match 0 at every n_{active} on the 2-cliques OOD set. The pairwise accuracy is in 0.74–0.77 across runs, and the per-distance breakdown shows a small but consistent ordering: the $D \leq 7$ and $D \leq 9$ models recognise the within-clique $d = 1$ pairs better than the unfiltered model (0.42 and 0.36 respectively, versus 0.08 for unfiltered and 0.0 for $D \leq 11$), at the cost of slightly worse accuracy on the "disconnected" across-clique pairs. The net effect on exact match is zero. We highlight this in passing because it is the only setting in which a filtered run has a sub-metric that exceeds the unfiltered model; the overall picture remains that the data lever does not produce a 2-cliques generalisation for the standard transformer at $n = 40$.

Unfiltered ER, by graph diameter

Model	$D \leq 7$	$D \leq 9$	$D \leq 11$	$D > 11$	overall
Unfiltered	1.000	0.9996	0.9991	**0.9986**	**0.9990**
$D \leq 11$	1.000	0.9996	0.9989	0.9807	0.9963
Curriculum	1.000	0.9996	0.9991	0.9704	0.9949
$D \leq 9$	1.000	**1.0000**	0.9883	0.5882	0.9302
$D \leq 7$	**1.000**	0.9277	0.7547	0.0324	0.6498

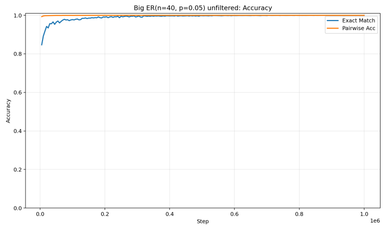
Table 5: Exact-match accuracy on a fixed test set of 10,000 unfiltered $\text{ER}(40, 0.05)$ graphs, broken down by the test-graph diameter, for the five large-model runs (number of graphs per bucket: 1,329, 5,574, 8,548, 1,452). Bold entries mark the best model in each column.

With every large-model run essentially perfect inside its training diameter budget, the table is a clean read of the data lever's actual effect. The unfiltered model is strictly best (or tied) on every diameter bucket and is overall the most accurate of the five. The filtered models are perfect on their training bucket and degrade monotonically on buckets above the cut-off, with the degradation becoming sharper the tighter the filter. The curriculum sits between $D \leq 11$ and unfiltered.

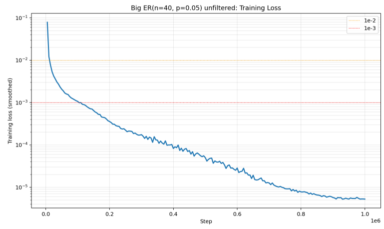
5.4 Per-experiment figures (large model)

The following subsections collect the in-distribution training curves and the per-experiment OOD plots used in the cross-experiment view above.

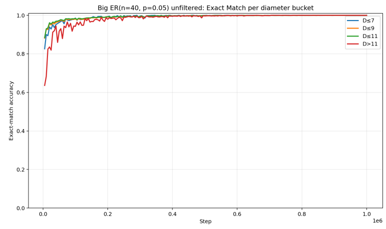
5.4.1 Unfiltered



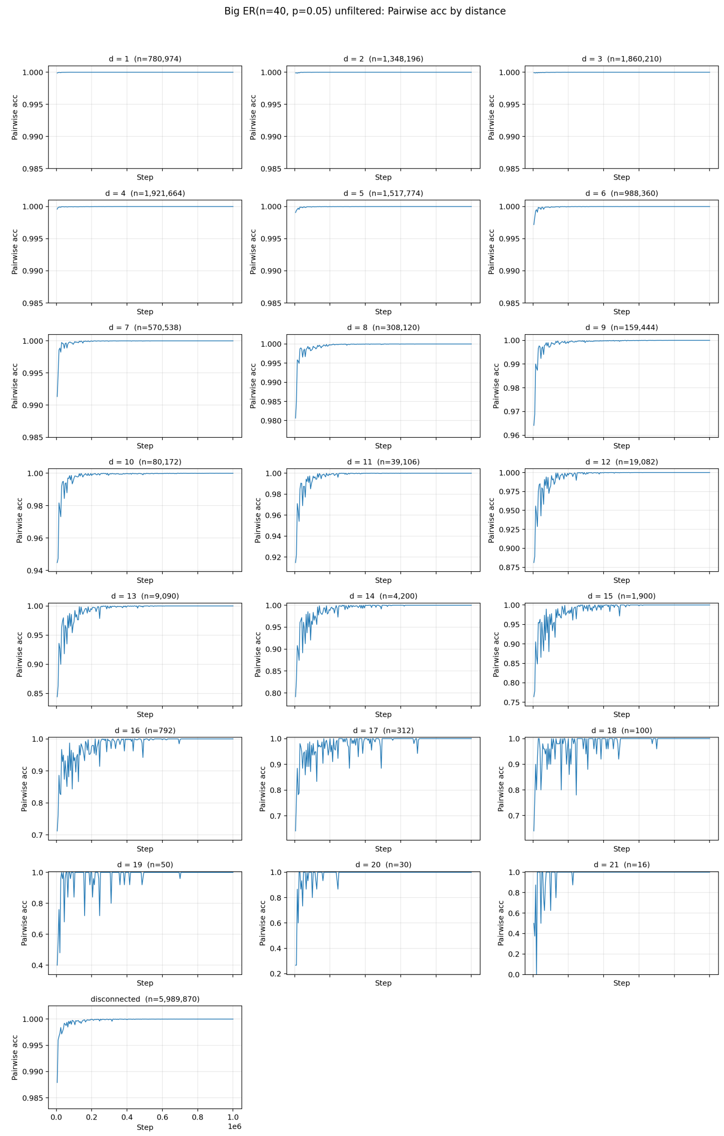
(a) Exact match and pairwise accuracy.



(b) Training loss.

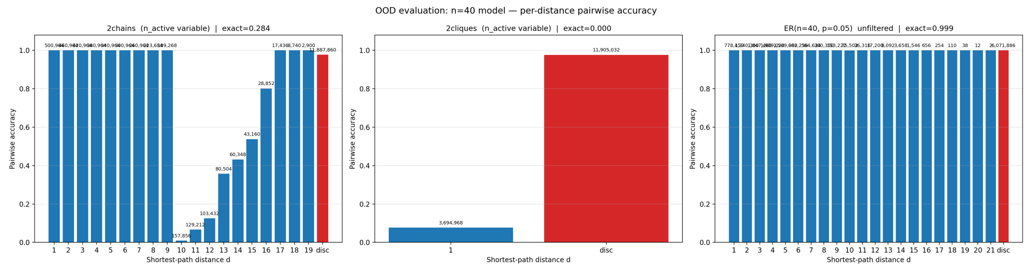


(c) Exact match per test-graph diameter bucket.

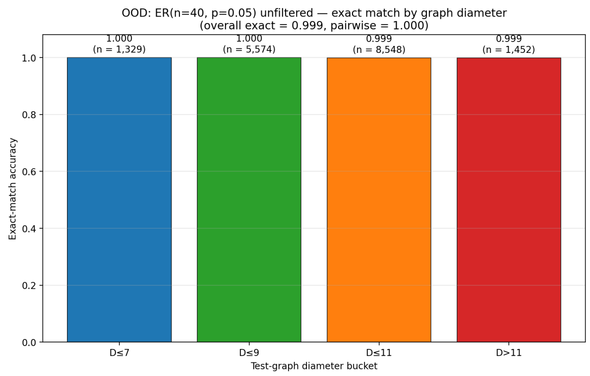


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

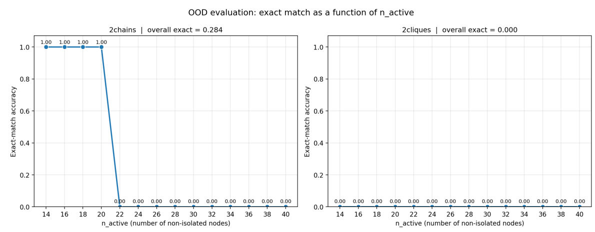
Figure 19: Large model, Unfiltered, in distribution.



(a) Per-distance pairwise accuracy on the three OOD sets.



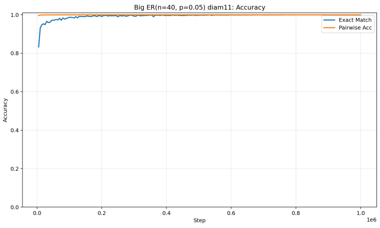
(b) Exact match on unfiltered ER by diameter bucket.



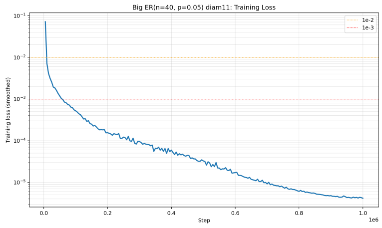
(c) Exact match on 2-chains and 2-cliques as a function of n_{active}.

Figure 20: Large model, Unfiltered, OOD.

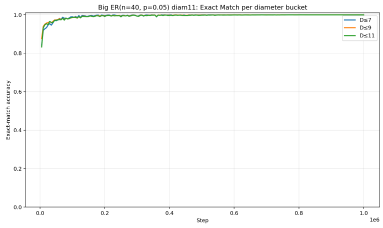
5.4.2 $D \leq 11$



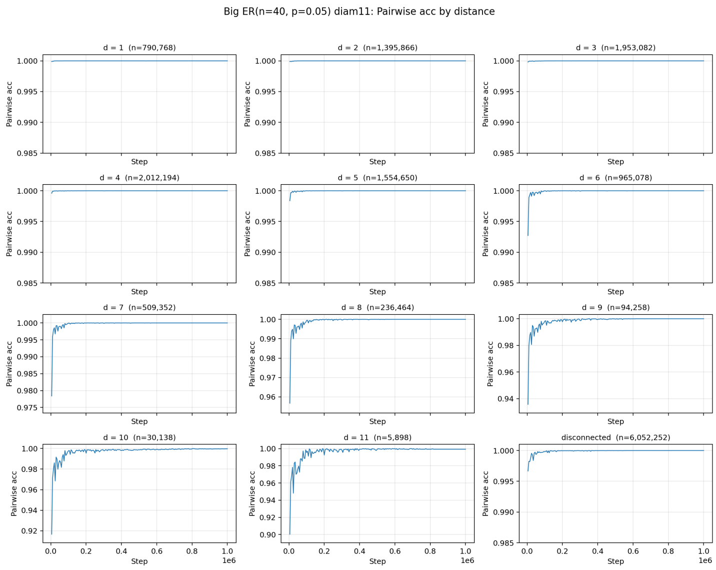
(a) Exact match and pairwise accuracy.



(b) Training loss.

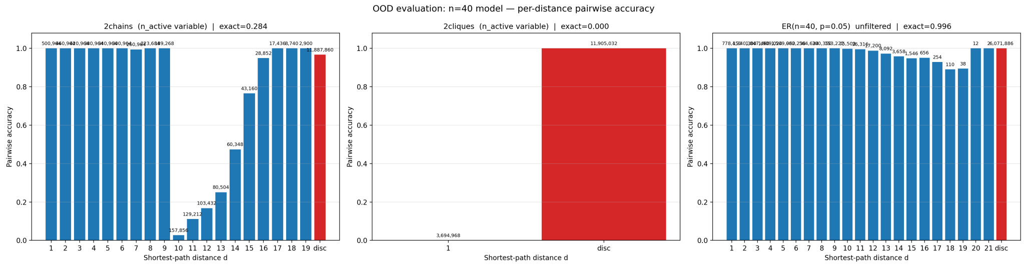


(c) Exact match per test-graph diameter bucket.

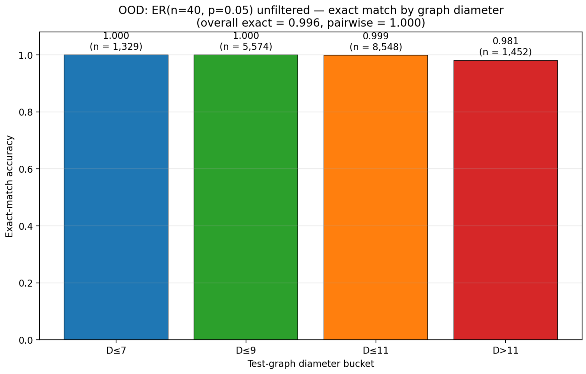


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

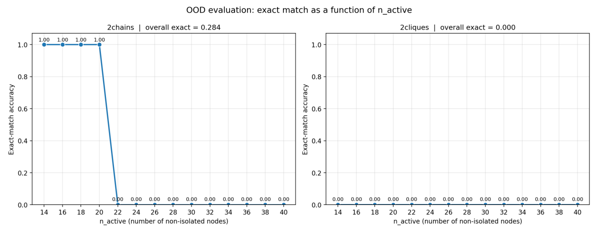
Figure 21: Large model, $D \leq 11$, in distribution.



(a) Per-distance pairwise accuracy on the three OOD sets.



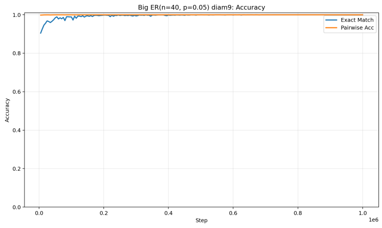
(b) Exact match on unfiltered ER by diameter bucket.



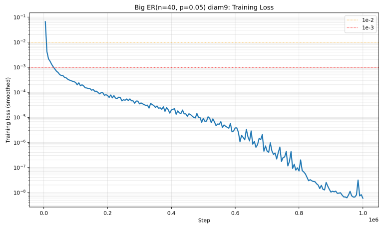
(c) Exact match on 2-chains and 2-cliques as a function of n_{active}.

Figure 22: Large model, $D \leq 11$, OOD.

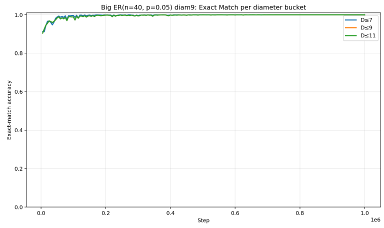
5.4.3 $D \leq 9$



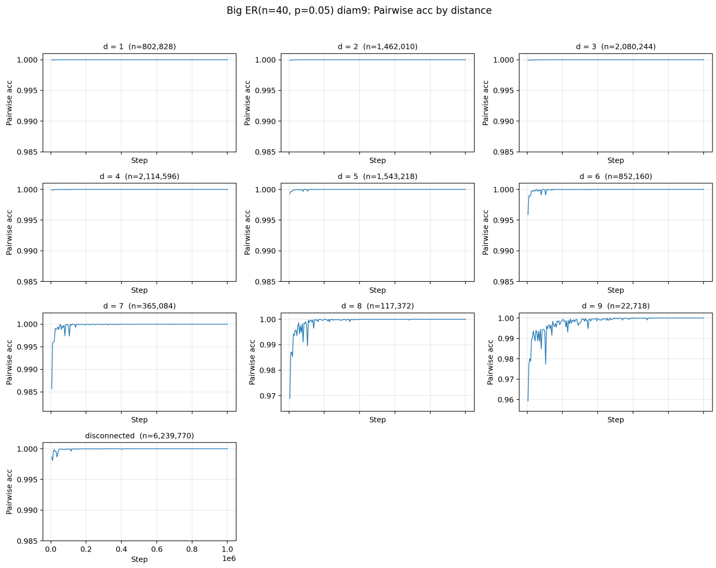
(a) Exact match and pairwise accuracy.



(b) Training loss.

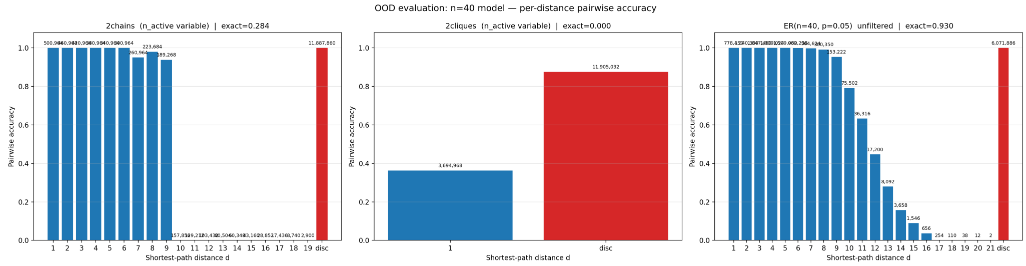


(c) Exact match per test-graph diameter bucket.

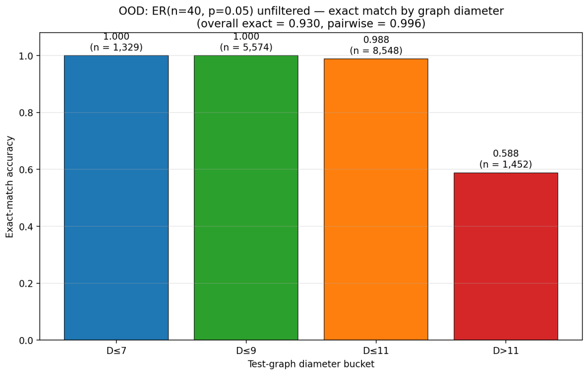


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

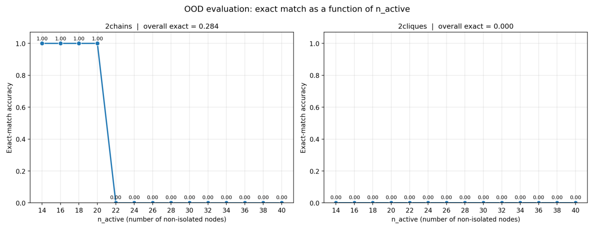
Figure 23: Large model, $D \leq 9$, in distribution.



(a) Per-distance pairwise accuracy on the three OOD sets.



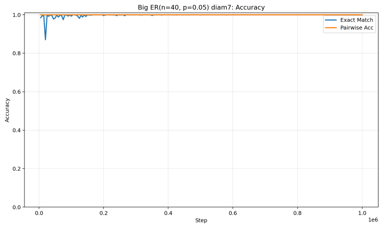
(b) Exact match on unfiltered ER by diameter bucket.



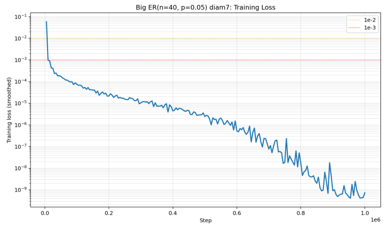
(c) Exact match on 2-chains and 2-cliques as a function of n_{active}.

Figure 24: Large model, $D \leq 9$, OOD.

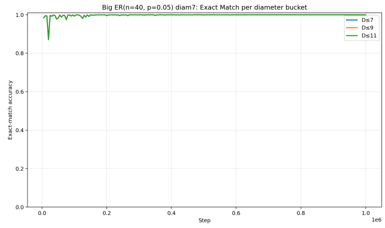
5.4.4 $D \leq 7$



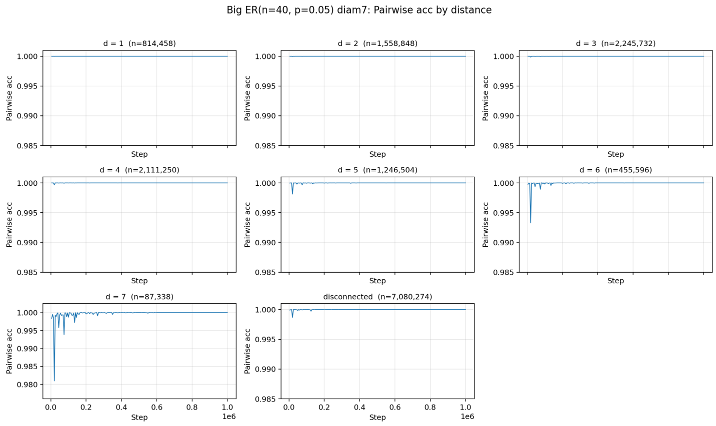
(a) Exact match and pairwise accuracy.



(b) Training loss.

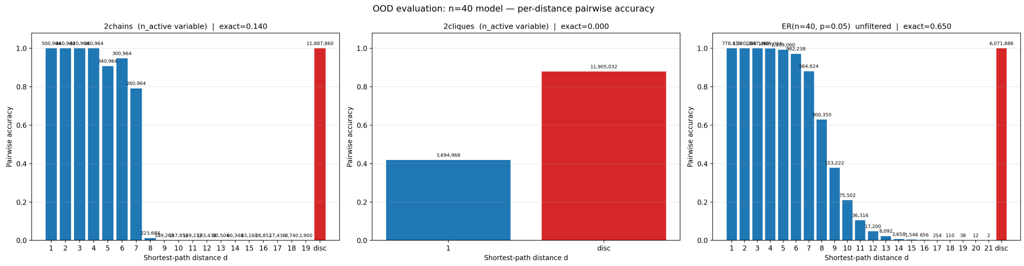


(c) Exact match per test-graph diameter bucket.

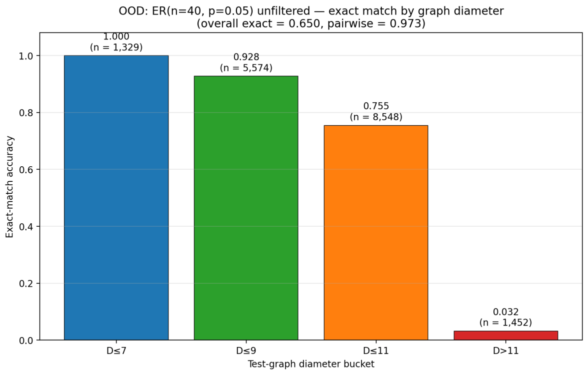


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

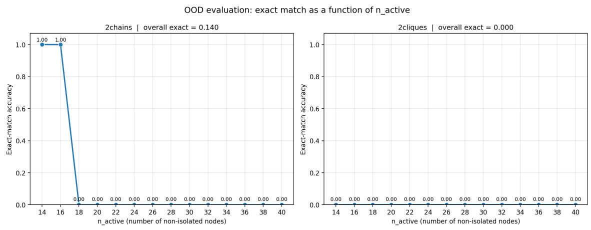
Figure 25: Large model, $D \leq 7$, in distribution.



(a) Per-distance pairwise accuracy on the three OOD sets.



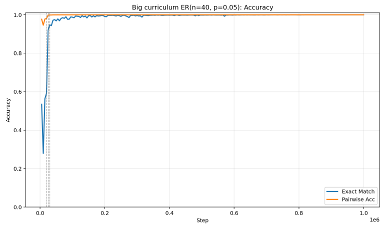
(b) Exact match on unfiltered ER by diameter bucket.



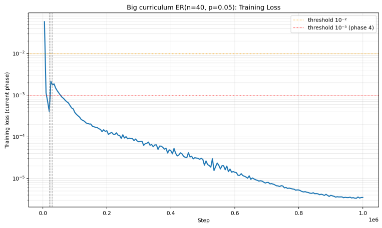
(c) Exact match on 2-chains and 2-cliques as a function of n_{active}.

Figure 26: Large model, $D \leq 7$, OOD.

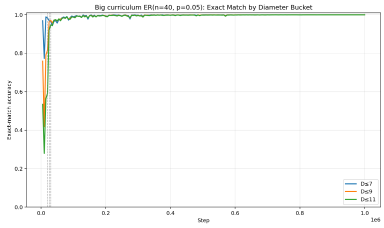
5.4.5 Curriculum



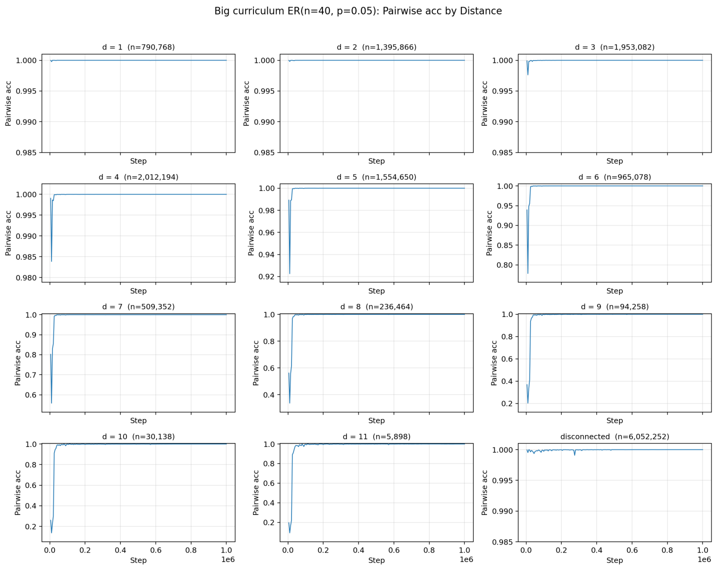
(a) Exact match and pairwise accuracy.



(b) Training loss.



(c) Exact match per test-graph diameter bucket.

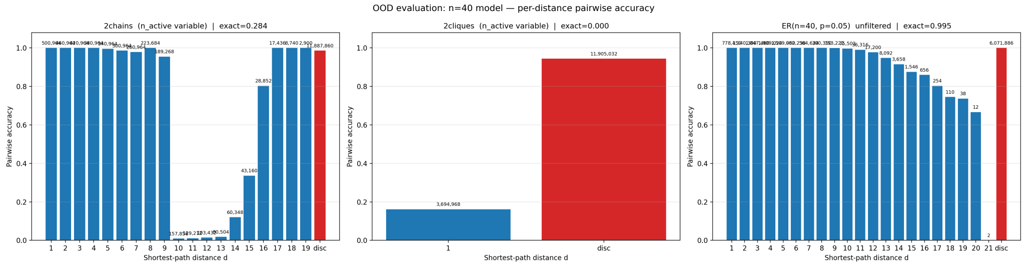


(d) Pairwise accuracy by shortest-path distance. The number above each panel is the count of test pairs at that distance.

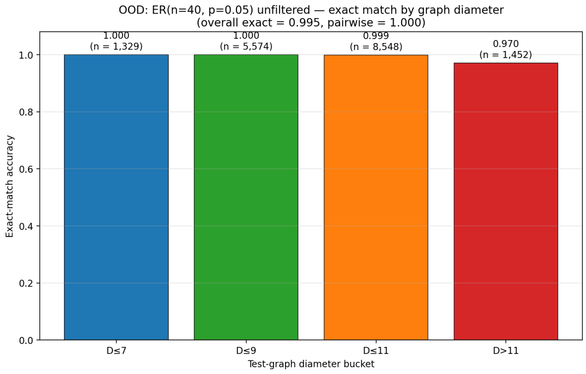
Figure 27: Large model, Curriculum, in distribution.

Phase (filter)	Start step
1 ($D \leq 7$)	0
2 ($D \leq 9$)	20,000
3 ($D \leq 10$)	25,000
4 ($D \leq 11$)	30,000

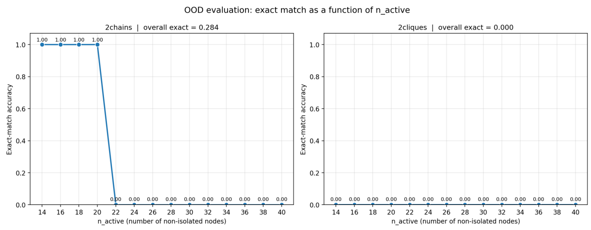
Table 6: Curriculum phase transitions. Phases 1–3 are advanced as soon as the smoothed training loss falls below a preset threshold; phase 4 runs to the end of the 1,000,000-step budget. Phases 1–3 close within the first 30,000 steps, so the remaining $\sim$ 970,000 steps are spent on $D \leq 11$.



(a) Per-distance pairwise accuracy on the three OOD sets.



(b) Exact match on unfiltered ER by diameter bucket.



(c) Exact match on 2-chains and 2-cliques as a function of n_{active}.

Figure 28: Large model, Curriculum, OOD.

5.5 Comments on Chapter 5

Two observations summarise the large-model results.

First, raising the model from $\sim$ 72k to $\sim$ 6.3M parameters lifts the 0.80 in-distribution ceiling of Chapter 4. All five large-model runs reach ≥ 0.9995 in-distribution exact match, and the convergence ordering across diameter filters is the same we already saw in the small-model regime: tighter filter, faster convergence.

Second, none of the OOD tests we ran show a benefit of the diameter filter in the large-model regime either. The unfiltered model is strictly best (or tied) on every bucket of the per-diameter unfiltered-ER evaluation, the diameter-restricted models exhibit a sharp accuracy cliff just above their cut-off, the 2-chains exact-match cliff at $n_{\text{active}} = 22$ is the same for unfiltered, $D \leq 11$, $D \leq 9$ and curriculum (and is earlier for $D \leq 7$), and all five runs reach exact match 0 on 2-cliques. The only sub-metric on which a filtered run exceeds the unfiltered model is within-clique $d = 1$ pairwise accuracy ($D \leq 7$ at 0.42, $D \leq 9$ at 0.36, versus 0.08 for unfiltered), which does not translate into any improvement at the graph level. We do not draw a strong conclusion from this single sub-metric.

Combined with Chapter 4, the empirical picture is consistent between the two model scales: in our standard-transformer setup the data lever helps the model learn faster but does not produce the OOD benefits predicted by the disentangled-transformer analysis. Whether this discrepancy is due to the architectural simplification between the disentangled and standard transformers, to the difference in scale ($n = 40$ here versus $n = 20$ in the original experiments), or to some combination of both, is not something our experiments can directly settle.